Probabilistic and Financial Literacy Impact on Ambiguity Attitudes towards Financial Instruments*

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Abstract

We examine the impact of standard and probabilistic financial literacy on ambiguity attitudes towards real-world financial assets. In a laboratory experiment, we provide subjects with either no literacy, only standard financial literacy, or we couple this with probabilistic literacy - cognitive instructions highlighting the variability of asset prices. We then elicit Ambiguity attitudes for the stock of a locally well-known company and for Bitcoin. With respect to the no literacy group, we find that Probabilistic Literacy decreases ambiguity aversion towards both assets, with the effect concentrated among those who deemed the treatment most useful. Probabilistic Literacy increases (decreases) ambiguity insensitivity towards the stock (Bitcoin). Standard Financial Literacy does not move ambiguity attitudes for any assets. Probabilistic Literacy also reduces choice noise relative to both other treatment arms. These findings underscore the importance of probabilistic reasoning in financial education and show that ambiguity attitudes - and their responsiveness to literacy interventions - depend critically on the underlying financial asset.

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JEL Codes: D81, G40, G53.

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1 Introduction

Investing in financial assets is a prominent example of choice under ambiguity: the future distribution of returns is unknown to any investor (Trautmann and Van De Kuilen, 2015). Ambiguity attitudes naturally vary across investors, but the same investor may also change her attitudes depending on her circumstances (Fox and Tversky, 1995), or hold different attitudes towards different assets (Anantanasuwong et al., 2024). At the same time, abundant research has highlighted the relevance of financial literacy for households' financial decisions (Lusardi and Mitchell, 2014). In our work, we try to understand the impact of different types of financial literacy on ambiguity attitudes towards different financial assets. In particular, we focus on the stock and the cryptocurrency market. On the one hand, limited stock market participation is observed worldwide (Gomes et al., 2021; Henkel and Pugnaghi-Zimpelmann, 2026; Veld et al., 2026). On the other hand, cryptocurrencies have spread rapidly, with high early adoption among the least financially literate investors (Panos et al., 2020). While institutional and behavioral factors are certainly at play, these contrasting patterns make stocks and cryptocurrencies a particularly informative setting for examining whether financial literacy shapes perceptions of uncertainty across asset classes.

In this paper, we study the impact of different types of financial literacy on ambiguity attitudes towards stocks and cryptocurrencies. We hypothesize that financial literacy shapes ambiguity attitudes of investors, with potentially different effects across asset classes. To do so, we run an experiment where we test two different types of financial literacy and elicit ambiguity attitudes towards the stock of a locally well-known company (Local Stock) and towards Bitcoin. Subjects received either i) a Standard Financial Literacy training, where they learned notions about financial instruments; ii) a Probabilistic Literacy Training, where the previous treatment was integrated with cognitive instructions meant to make salient that prices of financial instruments can vary in all directions; iii) no Literacy. After this phase, we elicited ambiguity attitudes through matching probabilities, following Baillon et al. (2018). We derived ambiguity attitudes employing a random-coefficient Hierarchical Bayes procedure (Gutierrez and Kemel, 2024), which recovers ambiguity attitudes reliably even when subjects make occasional mistakes. We derived i) ambiguity aversion, which measures the overall reluctance to bet on an ambiguous source with unknown probabilities, with respect to risky events with known probabilities; ii) ambiguity insensitivity, which captures the extent to which the subject blurs the distinction between more and less likely events. The two indices capture two different aspects of attitudes towards uncertain risk. The first one captures how much a subject (dis)likes risk with unknown probabilities with respect to risk with known probabilities. The second one captures how much uncertainty the subject perceives when facing risk with unknown probabilities, with the highest insensitivity meaning that different states of the world are perceived as equally likely, hence leading to overweighting unlikely events and underweighting likely ones.

We expect Financial Literacy to have an impact on attitudes towards uncertainty for two reasons: familiarity and reasoning. Under the comparative ignorance hypothesis (Fox and Tversky, 1995), ambiguity aversion is driven by the comparison with more familiar events. Hence, we suggest that increased financial literacy might increase familiarity with the assets and reduce ambiguity aversion. However, it is also known that ambiguity aversion increases when people are provided with diagnostic information that they do not know how to use (Fox and Weber, 2002). Hence, we do not make a clear prediction on how our intervention will modify ambiguity aversion. Moreover, we do provide two types of literacy. Probabilistic literacy is meant to help cognitive reasoning (Oprea and Woodford, 2026) and potentially reduce complexity (Oprea, 2024), although an increase is also possible. The ability to reason through probabilistic states should influence ambiguity generated insensitivity. The direction of this influence is not clear. In principle, teaching probabilistic reasoning should decrease insensitivity, but it can also lead to further reflections on likelihoods of different states and lead to an increase. The effect does not need to be constant across assets, as probabilistic reasoning should have a differential effect in different contexts.

We find that, with respect to the Control group, the Probabilistic Literacy Treatment decreases ambiguity aversion towards both the Local Stock and Bitcoin, but only for those participants who rate the treatment as moderately to very useful. It also increases ambiguity insensitivity towards the Local Stock, and it decreases it towards Bitcoin. The Standard Financial Literacy Treatment does not change any ambiguity attitude with respect to the Control. Also, we observe that the Probabilistic Literacy treatment leads to lower noise in choices than both the Financial Literacy Treatment and the Control. Our results are relevant for understanding which types of financial literacy interventions may shape households' ambiguity attitudes and, potentially, their investment decisions. Atkinson and Messy (2012) define financial literacy as "a combination of awareness, knowledge, skill, attitude and behavior necessary to make sound financial decisions and ultimately achieve individual financial wellbeing". We argue that probabilistic reasoning is a core component of financial literacy when the goal is to shape attitudes towards uncertainty in financial markets, complementing standard financial literacy. In this sense, our findings underscore the importance of *awareness, attitude and behavior* on top of the foundational components of *knowledge and skill*. In our sample, Probabilistic Literacy reduces ambiguity aversion, making the uncertainty associated with holding financial assets more acceptable. It also reduces decision noise, hence improving decision quality.

The results for ambiguity insensitivity are less straightforward. Relative to the Control group, the Probabilistic Literacy treatment increases ambiguity insensitivity towards the Local Stock: subjects become less likely to distinguish between more and less likely events. For Bitcoin, by contrast, the treatment reduces ambiguity insensitivity, leading subjects to discriminate more clearly between events with different likelihoods. Thus, the same probabilistic training has asset-specific effects on how subjects perceive the relative likelihood of possible outcomes. One possible explanation is that prompting subjects to distinguish

between more and less likely events increases the influence of extreme outcomes in memory, and that the most salient outcomes differ across assets (Bordalo et al., 2012, 2020). For the Local Stock, crashes may be particularly salient, making subjects think that their understanding of the distribution of possible outcomes is more limited than they had thought, resulting in greater ambiguity insensitivity. For Bitcoin, instead, extreme price increases may be more salient[1], making differences across outcomes appear more pronounced and thereby reducing ambiguity insensitivity.

We inform the literature on the study of ambiguity attitudes. Ambiguity attitudes are relevant because, in real life, we mostly face situations with unknown probabilities. This highlights the need to more often elicit ambiguity towards natural events, rather than focusing solely on artificial ones (Baillon et al., 2018). In general, people are ambiguity averse, preferring risky bets to ambiguous ones (Ellsberg, 1961; Halevy, 2007; Trautmann and Van De Kuilen, 2015). Ambiguity aversion has predictive power for real-life decisions (Bianchi and Tallon, 2019; Dimmock et al., 2016a,b; Ghazi et al., 2025; Ross et al., 2012; Sutter et al., 2013; Thimme and Völkert, 2015; Warnick and Laszlo, 2011). In particular, Dimmock et al. (2016a) show that ambiguity aversion can explain low stock market participation and the tendency among US investors to not own stocks. Moreover, studies have shown that people's ambiguity attitudes vary across different sources (Abdellaoui et al., 2011; Anantanasuwong et al., 2024; Heath and Tversky, 1991; Li et al., 2018).[2] In our work, we show that ambiguity attitudes can differ across different types of assets and that they are influenced by probabilistic reasoning training.

We also contribute to the fast-growing literature on financial literacy (Kaiser and Lusardi, 2024). Abundant correlational evidence is available on how financial literacy relates to economic outcomes. Financial literacy is highly correlated with: i) activity in financial markets (Christelis et al., 2011, 2010; Van Rooij et al., 2011); ii) the likelihood of undertaking retirement planning (Lusardi and Mitchelli, 2007); iii) the ability to cope with negative macroeconomic shocks (Klapper et al., 2013); iv) higher returns on savings account (Deuflhard et al., 2019); v) higher returns on retirement plan investments (Clark et al., 2017). On the other hand, a lack of financial knowledge is correlated to inefficient financial behaviors. For example: i) underdiversification and less frequent rebalancing (Abreu and Mendes, 2010; Calvet et al., 2007, 2009; von Gaudecker, 2015); ii) higher mortgage costs (Moore, 2003); iii) likelihood of using high-cost methods of borrowing (Lusardi and de Bassa Scheresberg, 2013; Lusardi and Tufano, 2015). In the plethora of studies on the impact of financial literacy, those showing causal evidence of it are a minority, possibly because of endogeneity issues. This literature mainly relies on randomized field experiments, and the evidence is mixed. Some show that financial literacy can: i) have an impact on

[1]The experiment was run in late 2020. The price of Bitcoin was on an upward trajectory and, although it had experienced some severe downturns, these were not as numerous as they later became.

[2]Several sources of uncertainty, such as temperature, language, and stock market returns, have been studied (Abdellaoui et al., 2011; Baillon et al., 2018; Chew et al., 2012; Heath and Tversky, 1991; Li, 2017).

saving and borrowing decisions (Haliassos et al., 2020; Sayinzoga et al., 2016); ii) boost retirement planning (Duflo and Saez, 2003; Song, 2020); iii) improve business practices (Drexler et al., 2014); iv) improve intertemporal decision-making (Lührmann et al., 2018); v) lead to higher accumulation of wealth (Bernheim et al., 2001). However, others find no significant changes in the financial behavior of subjects exposed to financial literacy programs. See, for example Carpena et al. (2019); Choi et al. (2011); Collins (2013) and Hastings et al. (2013) for a review. Also, mixed results are possible. For example, Brown et al. (2016) study the effects of exposure to financial training on debt outcomes in early adulthood. They find that mathematics and financial education decrease reliance on non-student debt and improve repayment behavior. Economics training, on the other hand, increases both the likelihood of holding outstanding debt and the prevalence of repayment difficulties. Talking to this literature, we contribute causal evidence that probabilistic reasoning moves ambiguity attitudes towards financial assets while standard financial literacy does not.

Our closest predecessor is Nieddu and Pandolfi (2021). In a laboratory experiment, they observe a gap in the certainty equivalent of a risky lottery and an equivalent risk task framed as a financial investment. Providing financial training eliminates this gap. They suggest their finding can be rationalized through the lens of ambiguity aversion. Similar to them, we provide financial literacy to a subset of our participants. They do find an effect of standard financial literacy, where they explain basic financial concepts, coupled with some basic numeracy concepts on how to calculate returns. In our probabilistic training, we provide cognitive instructions meant to highlight possible variations in the price of financial instruments. Unlike them, we can directly test whether financial or probabilistic training shifts ambiguity attitudes, and we show that financial training does not, while probabilistic training decreases ambiguity aversion towards a Local Stock and Bitcoin, and it increases (decreases) ambiguity insensitivity towards the stock (Bitcoin). Our results highlight that ambiguity attitudes can be moved by more complete probabilistic reasoning, rather than simpler financial literacy, and effects are heterogeneous across assets.

2 Ambiguity for Natural Events

This section sets out the conceptual and econometric framework underlying our analysis. We begin by defining the matching probabilities used to elicit choices over events. We then introduce the source-theoretic model, which separates subjective beliefs from ambiguity attitudes and summarizes the latter with two indices: ambiguity aversion and a-insensitivity. Finally, we describe the random-coefficient Hierarchical Bayes procedure used to estimate these components while accounting for individual heterogeneity and occasional choice errors.

2.1 Eliciting Matching Probabilities

Formally speaking, ambiguity does not concern just a single event E, but a partition, or, more generally, a source of uncertainty. Following Baillon et al. (2018), we assume a minimal degree of richness of the sources of uncertainty considered. There should be three mutually exclusive and exhaustive non-null events: E_1, E_2, and E_3, which in our case refer to the returns of the assets for which we elicit ambiguity attitudes. In order to estimate ambiguity attitudes towards a given asset, we derive matching probabilities for the six aforementioned events (E_1, E_2, and E_3 and their complements) defined on the space of the returns of the asset. We follow the definition of Baillon et al. (2018) for matching probabilities: receiving an amount X under event E is equivalent to receiving X with probability m. We elicit matching probabilities using multiple-price lists, as we later explain in Section 3. Estimating ambiguity attitudes from the collected data raises two main challenges. First, the elicited matching probabilities do not directly reflect ambiguity attitudes, and isolating the attitude component requires a theoretical framework. Second, even with the attitude component isolated, individual-level estimates can be sensitive to subjects' occasional mistakes given the limited observations. We now explain how we can address both challenges. In Section 2.2, we build on source theory (Abdellaoui et al., 2011) to precisely isolate ambiguity attitudes and express them with two independent indices, ambiguity aversion and a-insensitivity. In Section 2.3, we introduce our main estimation method: a random-coefficient Hierarchical Bayes procedure (Gutierrez and Kemel, 2024), which recovers ambiguity attitudes reliably even when subjects make occasional mistakes.

2.2 Ambiguity model

In our experiment, subjects report matching probabilities for a set of ambiguous events. These probabilities do not, by themselves, identify ambiguity attitudes, because they reflect both subjective beliefs about the likelihood of the event and preferences towards ambiguity. The latter can shift matching probabilities away from beliefs. For example, a high matching probability for an event in which a stock's price increases may reflect either an optimistic belief that the event is likely or a preference for betting on the ambiguous event rather than on a lottery with known probabilities—that is, ambiguity seeking. Matching probabilities alone therefore cannot distinguish between subjective beliefs and ambiguity attitudes.

Hence, we need to model how the belief and the ambiguity attitudes jointly determine the matching probability. Following source theory (Abdellaoui et al., 2011; Baillon et al., 2021, 2025, 2018), we assume that subjects hold subjective probabilities even under ambiguity, similar to objective probabilities under risk. What sets ambiguity apart is that subjects evaluate these subjective probabilities with a weighting function that depends on the source, and this function represents the subject's ambiguity attitude. We adopt the neo-additive model (Chateauneuf et al., 2007), in which the weighting function is linear. Let $\pi(E)$ denote the subjective probability the subject assigns to event E under a given

source, and let τ_0 and τ_1 be the intercept and slope of the linear weighting function. The matching probability can thus be expressed as:

$$(1) \qquad m^*(E) \;=\; \begin{cases} 0 & \text{if } \pi(E) = 0, \\ \tau_0 + \tau_1\,\pi(E) & \text{if } 0 < \pi(E) < 1, \\ 1 & \text{if } \pi(E) = 1, \end{cases} \qquad \tau_0 \geq 0,\ \tau_1 \geq 0,\ \tau_0 + \tau_1 \leq 1.$$

As Equation (1) shows, whenever the subject is neither certain that the event will happen nor certain that it will not, the matching probability is a linear transformation of the subjective probability. An ambiguity-neutral subject, who is indifferent between dealing with ambiguity and risk, has $m^*(E) = \pi(E)$ for every event. Any departure of $m^*(E)$ from $\pi(E)$ therefore reflects non-neutral ambiguity attitudes, characterized by the two parameters (τ_0, τ_1).

Following the convention in the literature (Li et al., 2018; von Gaudecker et al., 2025), we reparametrize (τ_0, τ_1) into two indices that capture independent dimensions of ambiguity attitudes, ambiguity aversion (the b-index) and ambiguity-generated insensitivity, a-insensitivity (the a-index):

$$(2) \qquad a \;=\; 1 - \tau_1 \in [0, 1], \qquad b \;=\; 1 - \tau_1 - 2\tau_0 \in [-1, 1].$$

Ambiguity aversion b measures the overall reluctance to bet on the ambiguous source, with unknown probabilities, relative to risky events with known probabilities. A positive b means the subject is averse to the ambiguous source, and a larger b reflects stronger aversion, whereas a negative b means the subject is instead ambiguity seeking toward this source. The other index, a-insensitivity a, captures the extent to which the subject blurs the distinction between more and less likely events. A subject with $a > 0$ acts as if compressing all likelihoods toward fifty-fifty, overweighting unlikely events and underweighting likely ones. An a-index close to zero means the subject distinguishes between events of different likelihood almost as sharply as under risk. A value further from 0 denotes higher compression towards fifty-fifty.

As suggested by source theory, ambiguity attitudes can differ across sources. Previous studies document that the same subject holds substantially different attitudes toward different sources (Abdellaoui et al., 2011; Anantanasuwong et al., 2024; Gutierrez and Kemel, 2024; Li et al., 2018). We elicit ambiguity attitudes toward two separate sources: the Local Stock and Bitcoin. In the analysis, we therefore let both the subjective beliefs and the two attitude indices be source-specific, and ask whether the treatments change the ambiguity attitudes toward each source.

2.3 Estimation method

Recovering ambiguity attitudes is demanding, as several parameters must be estimated from each subject's limited observed choices. As a result, each choice carries substantial

weight, and even a single mistake can move an individual estimate substantially. This problem persists even when the estimation explicitly allows for such mistakes through a noise scale (Murphy and ten Brincke, 2018).

To resolve this, we estimate the model structurally with Hierarchical Bayes (Train, 2009). Hierarchical Bayes addresses the instability by pooling information across subjects. Specifically, it assumes that each subject's parameters are drawn from a joint distribution, and estimates all parameters at both the individual and the population level. This procedure allows subjects to differ in their behavior, while preventing the occasional mistake from driving any single subject's estimate to an unrealistic extreme. Previous work has confirmed the strength of this approach relative to alternative methods (Baillon et al., 2020; Murphy and ten Brincke, 2018; Nilsson et al., 2011), and several recent papers have adopted it to estimate decision models under risk and ambiguity (Aydogan et al., 2025; Bland and Rosokha, 2021; Georgalos, 2019; Schmidt, 2025).

More specifically, for each subject and source, behavior is summarized by a parameter vector comprising the three subjective probabilities assigned to the return events, the two attitude indices a and b, and a noise scale σ capturing the stochasticity of the subject's choices. The likelihood links the observed choices to the matching probabilities implied by the model. Using Markov chain Monte Carlo (MCMC), we jointly estimate subject-specific parameters and the treatment-specific population distribution from which they are drawn. This hierarchical structure allows for individual heterogeneity while shrinking weakly identified estimates towards the population mean, thereby reducing the influence of occasional choice errors. Following the random-coefficient approach of Gutierrez and Kemel (2024), we estimate the model separately for each treatment and compare the resulting population means. This allows us to assess whether the treatments shift ambiguity attitudes towards the Local Stock and Bitcoin.

3 Experimental Design and Derivation of Indices

We next describe the experiment; the full instructions are provided in the Appendix. Ethical approval was granted by the University of Warwick. The experiment was programmed and conducted using oTree (Chen et al., 2016) during the last quarter of 2020 at LEEX-PUCP (Laboratorio de Economía Experimental—Pontificia Universidad Católica del Perú). Because of the COVID-19 pandemic, subjects participated online in small, supervised sessions. Ten sessions were conducted in total. We recruited 204 participants. The show-up fee was 5 Peruvian soles,[3] and the average bonus payment was 3.97 Peruvian soles. At the end of the experiment, we randomly selected one task for payment and paid subjects according to their choice in that task.[4]

[3]This corresponded to approximately US$1.35 at the time.

[4]The task was selected from the 12 ambiguity tasks and 4 risk tasks described below.

3.1 Introductory Questions

At the outset of the experiment, all subjects answered a demographic survey and a financial questionnaire containing 20 questions. These questions spanned past financial experiences of the respondent, numeracy, and financial literacy. The full set of questions is reported in Appendix Section C.3.

3.2 Treatments

Subjects were then randomly allocated to three treatment arms:

- **Standard Financial Literacy** This group received this piece of information:

  Now you have the opportunity to earn an additional amount of money by choosing between different options in the following questions. Before making your decisions, we now give you information that could be relevant and that could help you make your decisions. Please read them carefully!

  1. **What is a financial asset?** A financial asset is a financial instrument that can be traded on financial markets. Examples include:
     - **Stocks**: these are a capital investment that represents partial ownership of a corporation and gives the right to a share of the profits and assets of that corporation. An example of stocks is the shares of Banco de Crédito del Perú.
     - **Cryptocurrency**: this is a new form of digital asset based on a network distributed across a large number of computers. This decentralized structure allows them to exist outside the control of governments and central authorities. An example of cryptocurrency is Bitcoins.
  2. **What is the price of a financial instrument?** The price of financial instruments reflects supply and demand factors in the market in which they operate. Therefore, over time, a particular financial asset may decrease in value, remain constant, or increase in value.

- **Probabilistic Literacy** This group received the same information of the Standard Financial Literacy group plus the following extra information:

  What are the chances that a financial instrument will increase or decrease in price over time? The probabilities of financial instruments increasing, remaining constant, or decreasing ultimately depend on the specific asset under consideration. **In particular, each probability is not necessarily equally likely.** For example, for a specific asset it might be more likely to increase in price, or vice versa (to drop

in price). As a result, when making investment decisions, individuals must be aware of potential changes in the price of a financial instrument and form predictions about the probability of each possibility (increase or decrease in price).

- **Control.** This group received no financial literacy information.

The Probabilistic Literacy treatment is designed to complement standard financial literacy by making the variability of asset prices salient and encouraging subjects to reason explicitly about the likelihood of possible price movements. Through this channel, it may affect both ambiguity aversion—the willingness to bet on prospects with unknown probabilities—and ambiguity insensitivity—the ability to distinguish between outcomes with different likelihoods. Because the most salient outcomes and the interpretation of probabilistic information may differ across assets, the effects of the treatment need not be the same for the Local Stock and Bitcoin.

3.3 Ambiguity Task

We then elicited ambiguity attitudes toward Bitcoin and Banco de Crédito del Perú (BCP). We chose these assets to distinguish attitudes toward a new and unconventional asset from those toward a mainstream financial asset, represented by the stock of one of Peru's largest and most widely recognized domestic banks. Following Baillon et al. (2018), we used a multiple-price list to elicit matching probabilities for six events defined over the past return distribution of each asset, similarly to Offerman et al. (2009). We used past daily returns from the entire 2018 and 2019 calendar years for both assets. We truthfully informed subjects that the events were based on past daily returns but did not disclose the period from which those returns were drawn. Subjects therefore faced a genuine decision under ambiguity. For both assets, the six events for which matching probabilities were elicited consisted of the following three events and their complements:

- $E_1 = (-\infty, 0)$
- $E_2 = [0, 0.02]$
- $E_3 = (0.02, +\infty)$

In a typical multiple-price list, we fixed the payoff from the left option, for example "You gain 15 Peruvian Soles (around USD 4.00) if event E_1 realizes". In the right column, we proposed the same reward with increasing probabilities, "You gain 15 Peruvian Soles with X% probability", with $X \in \{0, 0.025, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.975, 1\}$. The midpoint between the switching rows, the last one where the subject preferred betting on the event E_1 and the first one where the subject preferred betting on the known probability, was taken as the matching probability for E_1. Hence, for example, if a subject preferred betting on E_1 when the known probability was up to 0.6 and then he preferred

the known risk when the known probability was 0.7 or higher, we took 0.65 as the matching probability for E_1 for that subject. We repeated this procedure for all six events and imposed a single-switching requirement. The matching probabilities elicited in this phase are used to construct the ambiguity indices described in Section 2.2. If an ambiguity choice was randomly chosen to be incentivized, we randomly selected one row of the relevant list and paid subjects according to their choice. If a subject chose "You gain 15 Peruvian soles (around USD 5.00) if event E_1 realizes," they received 15 Peruvian soles if the event occurred. If they chose the lottery, they received 15 Peruvian soles with the probability specified in the selected row.

3.4 Final Questions

After the ambiguity tasks, we elicited risk attitudes. For each subject we measured the certainty equivalent $CE(p)$ of a prospect paying $X = 15$ Peruvian Soles with known probability p, for $p \in \{0.17, 0.33, 0.50, 0.83\}$. Participants chose either to take part in a lottery which paid 15 Peruvian Soles with probability p or a certain payment X that increased across rows with $X \in \{0, 1, 2, 3, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 9, 10, 11, 12.5, 15\}$. If a risky choice was chosen to be incentivized, we randomly selected one row from the corresponding list and implemented the subject's choice. If the subject chose the lottery, payment was determined by its realized outcome; if they chose the certainty equivalent, they received the corresponding sure amount.[5]

Finally, subjects assigned to the Standard Financial Literacy treatment rated the usefulness of the information they had received, while those assigned to the Probabilistic Literacy treatment rated the usefulness of the probabilistic component—the material specific to that treatment. Subjects in all three treatment arms then answered a few brief questions on financial literacy. Summary statistics for the experiment are reported in Table A1.

4 Results

The Probabilistic Literacy (PL) treatment, which provides both the basic financial information and a probabilistic-reasoning prompt, makes subjects significantly more willing to bet on both the familiar Local Stock and the unfamiliar Bitcoin. In contrast, the Standard Financial Literacy (FL) treatment, which provides information alone, produces no significant change. In Section 4.1, we show the main results of the Probabilistic Literacy treatment and ask for whom the intervention works best. In Section 4.2, we report the results of the Standard Financial Literacy treatment and its comparison with Probabilistic Literacy. Table A2 reports hierarchical-Bayes population means of the subjective probabilities across conditions.

[5]Results on risk aversion are reported in Table A3.

4.1 Probabilistic Literacy

In the Probabilistic Literacy treatment, subjects receive basic financial information about stocks and cryptocurrencies as well as a probabilistic-reasoning prompt. Specifically, they are told that possible price movements need not be equally likely, encouraging them to evaluate each event according to its likelihood. Table 1 summarizes the estimated ambiguity attitudes towards the Local Stock and Bitcoin. We first examine the treatment's effect on the ambiguity-aversion index b, which captures a subject's reluctance to bet on an ambiguous source relative to a risky prospect with known probabilities. A higher b indicates stronger ambiguity aversion.

As reflected in Table 1, subjects in the Control treatment are on average close to ambiguity-neutral toward both sources ($b = 0.005$ on the Local Stock and $b = 0.026$ on Bitcoin, neither significantly different from zero).[6] The treatment pushes aversion into ambiguity-seeking territory on both sources: b falls to -0.257 on the Local Stock (-0.261 against Control, $p = 0.038$) and to -0.272 on Bitcoin (-0.298, $p = 0.020$). Subjects thus become more willing to bet on both sources, and the two effects are of similar size.

Result 1. *Probabilistic Literacy Treatment decreases ambiguity aversion towards both the Local Stock and Bitcoin.*

We next turn to the treatment effect on the a-insensitivity index a, which captures the extent to which a subject blurs the distinction between more and less likely events. A higher a means stronger compression of likelihoods toward fifty–fifty, that is, a weaker ability to tell more likely events from less likely ones. Unlike aversion, the treatment effect on the a-index is source-specific. The treatment raises insensitivity toward the Local Stock ($+0.509$, $p < 0.001$) and lowers it toward Bitcoin (-0.190, $p < 0.001$). Treated subjects thus distinguish less sharply among the possible events of the Local Stock, while they discriminate better among the possible events of Bitcoin.

Result 2. *Probabilistic Literacy Treatment increases ambiguity insensitivity towards the Local Stock, and it decreases it towards Bitcoin.*

Beyond ambiguity attitudes, our estimation identifies a second margin: the consistency of subjects' choices for each source. The Probabilistic Literacy treatment significantly reduces the estimated noise scale σ for both assets. For the Local Stock, σ falls from 0.437 in the Control group to 0.286 under Probabilistic Literacy (a difference of -0.151, $p = 0.012$); for Bitcoin, it falls from 0.388 to 0.273 (a difference of -0.115, $p = 0.019$). These reductions indicate that subjects' choices become more consistent and more closely aligned with their underlying preferences. We return to this result in Section 4.2.

Having established that the intervention reduces ambiguity aversion, we next examine whether this effect varies across participants. Specifically, we consider whether it differs

[6] Near-neutral average attitudes, with a sizeable share of ambiguity-seeking subjects, are common when ambiguity is elicited for natural sources such as stock performance (Anantanasuwong et al., 2024; Baillon et al., 2018; Henkel, 2024; Li et al., 2018).

Table 1: **Ambiguity attitudes: Probabilistic Literacy versus Control** The first two columns are hierarchical-Bayes population means, estimated separately for each treatment. The last column is the posterior mean contrast for each index with two-sided posterior p-value in parentheses. This is obtained by randomly drawing pairs from the two treatments' posterior distributions of the population means and taking the difference between PL and Control, which gives the posterior distribution of the difference Δ. The p-value in parentheses is calculated as $p = 2 \times \min\{\Pr(\Delta > 0), \Pr(\Delta < 0)\}$, where each probability is estimated by the corresponding share of posterior draws of Δ above or below zero. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

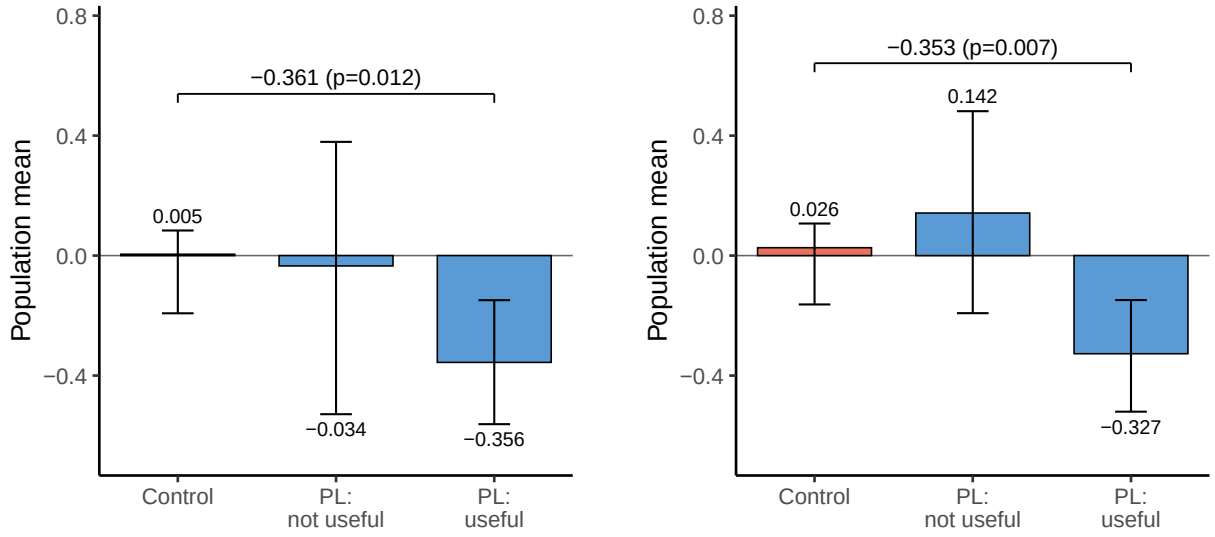
	Control	PL	PL − Control
Panel A: Local Stock			
b (aversion)	0.005	−0.257	-0.261^{**} (0.038)
a (insensitivity)	0.302	0.810	$+0.509^{***}$ (< 0.001)
σ (noise)	0.437	0.286	-0.151^{**} (0.012)
Panel B: Bitcoin			
b (aversion)	0.026	−0.272	-0.298^{**} (0.020)
a (insensitivity)	0.641	0.451	-0.190^{***} (< 0.001)
σ (noise)	0.388	0.273	-0.115^{**} (0.019)

according to participants' perceived usefulness of the intervention and their baseline financial literacy. Because these analyses involve estimating treatment effects in smaller subsamples, we focus on ambiguity aversion, both because it is the more policy-relevant index and because it yields more stable estimates under these sample splits. We begin with perceived usefulness.

At the end of the Probabilistic Literacy treatment, subjects rated how useful they found the information provided. We examine whether the treatment effect varies with this self-assessment by re-estimating the model using MCMC and allowing the population means of the ambiguity-aversion index to differ across usefulness groups. Usefulness was rated on a scale from 1 (not at all useful) to 5 (very useful). We divide treated participants into two groups: those who rated the treatment from 1 to 2 and those who rated it from 3 to 5. We then compare the estimated mean for each group with the Control estimate. Relative to the Control group, ambiguity aversion is lower by 0.361 for the Local Stock ($p = 0.012$) and by 0.353 for Bitcoin ($p = 0.007$) among participants who rated the treatment from 3 to 5. For participants who rated it from 1 to 2, the estimates do not differ significantly from the Control group for either asset (Figure 1). Thus, the reduction in ambiguity aversion is concentrated among participants who found the treatment useful. Corresponding results for a-insensitivity are reported in Figure A1.

Result 3. *Probabilistic Literacy Treatment decreases ambiguity aversion towards both the Local Stock and Bitcoin only for those participants who rate the treatment as moderately to very useful.*

We verify along which dimensions respondents who rated the treatment as useful differ from those who rated it as less useful. The usefulness rating is unrelated to gender, age, education, investment experience, and the objective literacy score. What distinguishes the respondents is their self-assessed knowledge: subjects who found the treatment useful rate their own knowledge of the Local Stock, of Bitcoin, and of finance in general higher than those who did not (Welch t-tests: $p = 0.026$, $p = 0.038$, and $p = 0.005$, respectively), even though the two groups do not differ significantly in objective literacy. This pattern suggests that participants who perceive the treatment as useful are those who feel confident—perhaps even overconfident—about their familiarity with the assets, rather than those with greater objectively measured knowledge. Responsiveness to the intervention is therefore associated with self-assessed, but not objective, knowledge. In real-world settings, the intervention may consequently be most likely to influence investors who are confident in their understanding of the assets.



(a) Local Stock

(b) Bitcoin

Figure 1: **The probabilistic-literacy effect on aversion by perceived usefulness** Bars represent hierarchical-Bayes population means of ambiguity aversion for the Control and for the Probabilistic Literacy subgroups that rated treatment usefulness from 1 to 2 vs. from 3 to 5, with 95% credible intervals. Bar labels report the posterior mean. Bracket labels report the pairwise contrast with its two-sided posterior p-value, shown when $p < 0.10$. This is obtained by randomly drawing pairs from the posterior distributions in each group and taking the difference between the mean in a PL subgroup by treatment usefulness and the Control, which gives the posterior distribution of the difference Δ. The p-value in parentheses is calculated as $p = 2 \times \min\{\Pr(\Delta > 0), \Pr(\Delta < 0)\}$, where each probability is estimated by the corresponding share of posterior draws of Δ above or below zero.

The second dimension of heterogeneity we examine is subjects' financial literacy before the intervention, measured as the number of correct answers on the financial literacy quiz. Although the literacy score does not correlate with whether subjects find the intervention useful, the effect itself may still vary with subjects' financial literacy, through a channel independent of the perception we just examined. We therefore re-run the MCMC allowing the treatment effect on ambiguity aversion to vary with the standardized literacy score. This yields an interaction term that captures whether and how the treatment effect changes as literacy increases. On the Local Stock, the interaction coefficient is $+0.236$ ($p = 0.092$), implying that the effect is larger among the less literate subjects. On Bitcoin, the interaction is essentially zero (-0.062, $p = 0.608$), implying that the change in ambiguity aversion is uniform across literacy levels. The intervention appears to help the less literate more, although the difference shows up only on the Local Stock and it is barely significant. This observed asymmetry might have an intuitive explanation: on the Local Stock, the intervention substitutes for the skill that the less literate lack, while on Bitcoin, nearly all subjects start as novices and benefit by similar magnitudes.

4.2 Standard Financial Literacy versus Probabilistic Literacy

We now turn to the Standard Financial Literacy treatment, where only the basic financial information is provided. Relative to the Control, Standard Financial Literacy leaves both aversion and insensitivity statistically unchanged on both assets (aversion -0.151 on the Local Stock, $p = 0.107$, and -0.155 on Bitcoin, $p = 0.136$, see Table 2). Nevertheless, the aversion estimates under Standard Financial Literacy seem to lie between Control and Probabilistic Literacy. The difference between Probabilistic Literacy and Standard Financial Literacy is not statistically significant either, and only the difference between Probabilistic Literacy and Control is, as shown in the previous Section. Overall, the results suggest a possible role for probabilistic reasoning beyond basic financial information.

Result 4. *We find no statistically significant effect of Standard Financial Literacy on ambiguity attitudes towards either the Local Stock or Bitcoin.*

Although the Probabilistic Literacy and Standard Financial Literacy treatments do not differ significantly in ambiguity aversion, they differ clearly in choice consistency. Under Standard Financial Literacy, the estimated noise scale is not significantly different from that under Control for either asset: the contrasts are $+0.018$ for the Local Stock ($p = 0.713$) and -0.008 for Bitcoin ($p = 0.848$). Under Probabilistic Literacy by contrast, the noise scale is significantly lower than under Standard Financial Literacy for both the Local Stock (-0.170, $p = 0.005$) and Bitcoin (-0.107, $p = 0.019$). These are reductions in noise of 37% and 28%, respectively between the Standard and the Probabilistic Financial Literacy.[7] Noise is also significantly lower under Probabilistic Financial Literacy than under Control.

[7] A reduction of noise tells us that subject more closely follow their own model-implied matching probability.

Table 2: **Ambiguity attitudes across conditions** The first two columns are hierarchical-Bayes population means, estimated separately for each treatment (the Control column repeats Table 1 for reference). The last two columns are the posterior mean contrasts for each index with two-sided posterior p-value in parentheses. These are obtained by randomly drawing pairs from the two treatments' posterior distributions of the population means and taking the difference between FL and Control (col. 3) or PL and FL (col. 4). In each case, this gives the posterior distribution of the difference Δ. The p-value in parentheses is calculated as $p = 2 \times \min\{\Pr(\Delta > 0), \Pr(\Delta < 0)\}$, where each probability is estimated by the corresponding share of posterior draws of Δ above or below zero. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Control	FL	FL − Control	PL − FL
Panel A: Local Stock				
b (aversion)	0.005	−0.147	−0.151 (0.107)	−0.110 (0.296)
a (insensitivity)	0.302	0.371	+0.070 (0.728)	$+0.439^{***}$ (< 0.001)
σ (noise)	0.437	0.456	+0.018 (0.713)	-0.170^{***} (0.005)
Panel B: Bitcoin				
b (aversion)	0.026	−0.129	−0.155 (0.136)	−0.143 (0.190)
a (insensitivity)	0.641	0.471	−0.170 (0.271)	−0.020 (0.778)
σ (noise)	0.388	0.380	−0.008 (0.848)	-0.107^{**} (0.019)

Since Probabilistic Literacy differs from Standard Financial Literacy only through the addition of the probabilistic-reasoning prompt, these results suggest that the reduction in noise is attributable to the prompt rather than to basic financial information. Lower noise indicates that choices are more closely aligned with subjects' estimated underlying preferences; it therefore reflects greater choice consistency and precision, rather than necessarily a normative improvement in decision quality. Thus, even a brief probabilistic prompt can materially improve choice consistency.

Result 5. *Probabilistic Literacy Treatment leads to lower noise in choices than both the Standard Financial Literacy Treatment and the Control.*

5 Discussion and Conclusion

In this work, we examined the effects of Standard Financial Literacy and Probabilistic Literacy training on ambiguity attitudes towards two financial assets: the stock of a locally well-known company and Bitcoin. We find no statistically significant effect of Standard Financial Literacy on ambiguity attitudes. Probabilistic Literacy, by contrast, has three distinct effects. First, it reduces ambiguity aversion towards both assets, with the effect concentrated among participants who rate the training as at least moderately useful. Second, it reduces noise in choices. Finally, it affects ambiguity insensitivity in opposite

directions across assets: it increases insensitivity towards the Local Stock while reducing it towards Bitcoin.

Several implications of our results are worth emphasizing. We contribute to the financial literacy literature by showing that standard financial literacy does not significantly alter ambiguity attitudes, whereas a more sophisticated form of probabilistic literacy does. This distinction has implications for the design of financial-education programs related to investment decisions. Innovative tools are being evaluated in this area, including narrative and visual tools (Lusardi et al., 2017) and gamified approaches (Freitas et al., 2025). Our findings suggest that probabilistic reasoning could also play a useful role. Moreover, Probabilistic Literacy reduces choice noise and ambiguity aversion, making subjects more willing to bet on ambiguous asset returns. We do not take a normative stance on whether greater willingness to invest is desirable. However, given widespread household aversion to stock investment, understanding the factors that shape this aversion remains important (Henkel and Pugnaghi-Zimpelmann, 2026).

Moreover, Probabilistic Literacy has opposite effects on ambiguity insensitivity across the two assets. It reduces subjects' ability to distinguish between more and less likely outcomes for the Local Stock, but improves this ability for Bitcoin. This result highlights the source dependence of ambiguity attitudes and may help inform the literature on why households invest differently in conventional financial assets and cryptocurrencies (Auer and Tercero-Lucas, 2022; Panos et al., 2020; Tercero-Lucas et al., 2026; Weber et al., 2023). One possible explanation is that probabilistic reasoning makes some past outcomes more salient in memory than others (Bordalo et al., 2012, 2020). For example, crashes may be particularly salient for the Local Stock, whereas extreme price increases may be more salient for Bitcoin. This interpretation is tentatively supported by Table A2: under Probabilistic Literacy, the estimated subjective probability assigned to a Bitcoin return above 2% increases by 14 percentage points relative to the Control group ($p < 0.001$), while the probability assigned to a negative return decreases by 9 percentage points ($p < 0.001$). We observe no comparable shifts for the Local Stock. Although the a-insensitivity index provides a more refined measure of discrimination among states than raw comparisons of subjective probabilities, these patterns are consistent with the proposed salience-based explanation. Further research on how direct and indirect experiences with different assets shape memories of past returns may help explain this divergence. The literature on the effects of financial memories on investment decisions is rapidly expanding (Brettschneider et al., 2026; Burro et al., 2026; Gödker et al., 2025; Jiang et al., 2025; Yeung et al., 2025). In particular, (Jiao and Li, 2021) observe, in a laboratory experiment, that a history of financial gains increases the likelihood of investing in an ambiguous asset.

Finally, we contribute to the literature on ambiguity attitudes. To our knowledge, we are among the first to study ambiguity attitudes towards financial assets. Our findings reinforce two conclusions: ambiguity attitudes are not stable individual traits (Gutierrez

and Kemel, 2024; Li et al., 2018), and they are strongly source-dependent (Gutierrez and Kemel, 2024). In particular, the opposite changes in ambiguity insensitivity across assets suggest that ambiguity attitudes vary not only across sources but also in their response to the same stimulus, depending on the source involved. Further research could investigate the origins of this heterogeneous response. One possibility is that the same cue makes different features more or less salient, thereby increasing or decreasing the decision weight assigned to those features depending on the context (Bordalo et al., 2026).

Declaration of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors used ChatGPT, Gemini, and Claude to proofread existing content and to speed up code writing in data analysis. In particular, experimental instructions were translated from Spanish using AI tools. After using these tools/services, the authors reviewed and edited the content as needed. The authors take full responsibility for the content of the publication.

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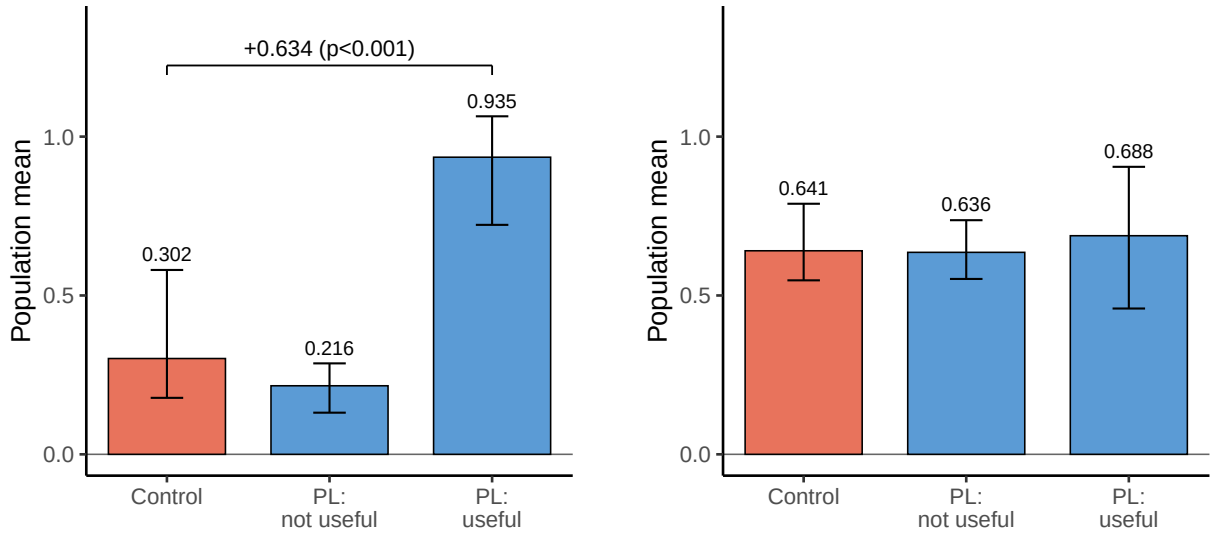
A Extra Tables

Table A1: **Summary Statistics by treatment** Each treatment is compared with Control via Welch's two-sided t-test.

Characteristic	Mean			PL − Control		FL − Control	
	Control ($n = 61$)	PL ($n = 66$)	FL ($n = 77$)	Δ	p	Δ	p
Male	0.39	0.38	0.40	−0.01	0.867	+0.01	0.914
Age (years)	22.79	22.98	21.95	+0.20	0.818	−0.84	0.105
Not in bachelor programme	0.18	0.18	0.10	+0.00	0.983	−0.08	0.211
Economics related or scientific faculty	0.64	0.64	0.70	−0.00	0.972	+0.06	0.447
Has taken a finance course	0.64	0.56	0.55	−0.08	0.369	−0.09	0.267
Self-rated finance knowledge (0–10)	4.13	3.52	4.49	−0.62	0.129	+0.36	0.353
Financial literacy score (0–15)	8.97	7.53	8.57	−1.44	0.021	−0.40	0.472
Local Stock knowledge (0–7)	3.51	3.09	3.78	−0.42	0.137	+0.27	0.326
Bitcoin knowledge (0–7)	2.54	2.50	2.90	−0.04	0.889	+0.36	0.222
Has ever invested	0.03	0.05	0.04	+0.01	0.715	+0.01	0.847

Table A2: **Subjective Probabilities by treatment** p_1, p_2, p_3 are the hierarchical-Bayes population means of the subjective probabilities of the three next-day return events of each source, from the same fits as Tables 1 and 2. The last two columns are the posterior mean contrasts for each probability with two-sided posterior p-value in parentheses. These are obtained by randomly drawing pairs from the two treatments' posterior distributions of the population means and taking the difference between PL and Control (col. 3) or FL and Control (col. 4). In each case, this gives the posterior distribution of the difference Δ. The p-value in parentheses is calculated as $p = 2 \times \min\{\Pr(\Delta > 0), \Pr(\Delta < 0)\}$, where each probability is estimated by the corresponding share of posterior draws of Δ above or below zero. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Belief	Mean			PL − Control	FL − Control
	Control	PL	FL		
Panel A: Local Stock					
p_1 (return $< 0\%$)	0.30	0.33	0.29	$+0.03^{***}$ (< 0.001)	−0.01 (0.647)
p_2 (return 0–2%)	0.36	0.27	0.37	-0.10^{***} (< 0.001)	+0.00 (0.838)
p_3 (return $> 2\%$)	0.33	0.40	0.34	$+0.06^{***}$ (< 0.001)	+0.00 (0.829)
Panel B: Bitcoin					
p_1 (return $< 0\%$)	0.34	0.25	0.28	-0.09^{***} (< 0.001)	-0.07^{**} (0.020)
p_2 (return 0–2%)	0.36	0.31	0.34	-0.06^{***} (0.003)	−0.02 (0.281)
p_3 (return $> 2\%$)	0.29	0.44	0.38	$+0.14^{***}$ (< 0.001)	$+0.08^{***}$ (0.001)



(a) Local Stock

(b) Bitcoin

Figure A1: **The probabilistic-literacy effect on a-insensitivity by perceived usefulness** Bars represent hierarchical-Bayes population means of a-insensitivity for the Control and for the Probabilistic Literacy subgroups that rated treatment usefulness from 1 to 2 vs. from 3 to 5, with 95% credible intervals. Bar labels report the posterior mean. Bracket labels report the pairwise contrast with its two-sided posterior p-value, shown when $p < 0.10$. This is obtained by randomly drawing pairs from the posterior distributions in each group and taking the difference between the mean in a PL subgroup by treatment usefulness and the Control, which gives the posterior distribution of the difference Δ. The p-value in parentheses is calculated as $p = 2 \times \min\{\Pr(\Delta > 0), \Pr(\Delta < 0)\}$, where each probability is estimated by the corresponding share of posterior draws of Δ above or below zero.

B Risk Aversion and Probability Weighting

As we explained in Section 3.4, we obtained the the certainty equivalent $CE(p)$ of a prospect paying $X = 15$ Peruvian Soles with known probability p, for $p \in \{0.17, 0.33, 0.50, 0.83\}$. Given the small stakes involved, we assume linear utility, so that the normalized certainty equivalent $CE(p)/X$ is a direct estimate of the probability weight $w(p)$. Following Abdellaoui et al. (2011), we summarize each source of uncertainty by a linear weighting function. For risk, we run an individual-level OLS regression of the normalized certainty equivalents on the four probabilities,

$$\frac{CE(p)}{X} = c + s\,p + \varepsilon, \tag{3}$$

and obtain a subject-specific intercept c and slope s. From these we compute two indices. The *insensitivity* index $a = 1 - s$ measures how much the weighting function flattens towards fifty-fifty: $a > 0$ means that unlikely outcomes are overweighted and likely ones underweighted, the inverse-S pattern documented by Wu and Gonzalez (1996) and Abdellaoui (2000). The *risk-aversion* index $b = 1 - s - 2c$ measures the elevation of the function, with $b > 0$ indicating that the prospect is valued below its expected value. These two indices are the known-risk counterparts of ambiguity aversion and ambiguity insensitivity indices.

We do observe an impact of our treatments on risk attitudes: Probabilistic Literacy lowers risk aversion significantly (-0.163, $p = 0.035$), while risk insensitivity is unchanged across treatments (Table A3). The pattern mirrors our results for ambiguity aversion: the Standard Financial Literacy estimate lies between Control and Probabilistic Literacy, and only the pairing of financial literacy with the probabilistic reasoning prompt moves risk aversion significantly.

Table A3: **Risk attitudes** Mean differences between treatments with Welch t-test p-value in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Comparison	Risk aversion	Risk insensitivity
PL − Control	-0.163^{**} (0.035)	$+0.063$ (0.522)
FL − Control	-0.118 (0.123)	$+0.133$ (0.179)
PL − FL	-0.045 (0.556)	-0.070 (0.442)

C Experimental Instructions

C.1 Welcome Page

Welcome to our experiment, and thank you for participating. The experiment will take about 20 minutes.

This research aims to understand how people make economic decisions. The study is conducted by researcher and student Alessandro Castagnetti of the University of Warwick (email: s.castagnetti@warwick.ac.uk).

For your participation today, we will pay you S/. 5.00. In addition, you can earn more money depending on the decisions you make in the experiment's activities. In each of these activities, you can receive up to a maximum of S/. 15.00. However, at the end of the experiment, the computer will randomly select (with equal probability) only one of these activities. The selected activity will determine your additional earnings from the experiment.

Please read the instructions for each part of the experiment carefully. All your decisions in the experiment are anonymous. This means that neither the other participants nor we will be able to link your decisions to your name in any way. The methodology of this study guarantees complete anonymity.

Please note that, as in all economic experiments, there is no deception in this experiment. In other words, all the instructions we give you are true. This research has been approved by the University of Warwick.

This experiment can only be completed on a computer. Completing the experiment on a tablet or mobile phone is not allowed.

If you have any questions during the experiment, you can contact us via WhatsApp at +51 941 475 026 or by email at leex@pucp.edu.pe.

To continue with the experiment, please click the "Next" button.

C.2 Demographics

Please answer these first questions.

- What is your sex?
  - Female
  - Male
  - Other
  - Prefer not to say
- What is your age? [Open numeric answer, between 18 and 100]
- What degree are you currently pursuing?

  - Bachelor's degree
  - Master's degree
  - PhD
  - Other/Not applicable

- What year of university are you in?
  - First year
  - Second year
  - Third year
  - Fourth year
  - Fifth year
  - Other/Not applicable

- Which faculty are you studying in?
  - Faculty of Science and Engineering
  - Faculty of Law
  - Faculty of Management and Senior Management
  - Faculty of Accounting
  - Faculty of Social Sciences
  - Faculty of Communication Sciences and Arts
  - Faculty of Arts and Humanities
  - Faculty of Psychology
  - Faculty of Art and Design
  - Faculty of Performing Arts
  - Faculty of Education
  - Faculty of Architecture and Urban Planning
  - Other
  - Not applicable

- Have you ever taken a finance course?
  - Yes
  - No

- On a scale from 0 to 10, how much would you say you know about finance (0 = nothing, 10 = expert)?
  - 0 (Nothing), 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 (Expert)

C.3 Questionnaire

Read each question and choose the correct answer. You will not be paid for your answers to these questions. In total, you have a maximum of 7 minutes to complete this questionnaire.

1. Have you ever invested in the stock market?
   - Yes, I invest frequently
   - Yes, I have invested at least once
   - No, never
2. On a scale from 0 to 7, how much do you know about the company "Banco de Crédito del Perú"?
   - 0 (I don't know it at all), 1, 2, 3, 4, 5, 6, 7 (I know it very well)
3. Have you ever bought shares of Banco de Crédito del Perú?
   - Yes
   - No, never
4. On a scale from 0 to 7, how much do you know about "Bitcoin"?
   - 0 (I don't know it at all), 1, 2, 3, 4, 5, 6, 7 (I know it very well)
5. Have you ever bought Bitcoin?
   - Yes
   - No, never
6. Suppose you have S/. 100 in a savings account and the interest rate is 2% per year. How much would you have in this account after 5 years?
   - More than 102
   - Exactly 102
   - Less than 102
   - I don't know
7. Imagine that the interest rate on your savings account is 1% per year and inflation is 2% per year. After 1 year, how much would you be able to buy with the money in this account?
   - More than today
   - The same as today

- Less than today
- I don't know

8. Do you think the following statement is true or false? Buying shares of a single company usually provides a safer return than a stock mutual fund.

   - True
   - False
   - I don't know

9. What is a stock?

   - A certificate of ownership of part of a company
   - A loan to a company
   - The obligation to pay a company's debts
   - None of the above
   - I don't know

10. What is a Bitcoin?

    - A new form of digital asset corresponding to ownership of a company
    - A new form of digital asset based on a network distributed across a large number of computers
    - A new form of digital asset that can only be used in Europe
    - None of the above
    - I don't know

11. When an investor spreads their money among different assets, the risk of losing money:

    - Increases
    - Decreases
    - Stays the same
    - I don't know

12. Stocks are usually riskier than bonds. True or false?

    - True
    - False
    - I don't know

13. If interest rates fall, what should happen to bond prices?

    - They increase
    - They decrease
    - They stay the same
    - I don't know

14. A 15-year mortgage typically requires higher monthly payments than a 30-year mortgage, but the total interest paid over the life of the loan will be less.

    - True
    - False
    - I don't know

15. When you buy a call option on a stock, you are actually buying:

    - The right to sell a stock in the future at a price set today
    - The right to buy a stock in the future at a price set today
    - The obligation to sell a stock in the future at a price set today
    - The obligation to buy a stock in the future at a price set today
    - I don't know

16. Someone gives you a SCRATCH AND WIN card that lets you win: S/. 10 with 50% probability, S/. 16 with 25% probability, and S/. 20 with 25% probability. Calculate and state the expected payoff. If you don't know, write "1234". Example: if your answer is 10, write: "10". [Open numeric answer]

17. Suppose a friend inherits S/. 10,000 today and his brother inherits S/. 10,000 three years from now. Who became richer from the inheritance?

    - Your friend
    - Your friend's brother
    - Both equally
    - I don't know

18. Suppose that in 2021 your income has doubled and the prices of all goods have also doubled. In 2021, how much will you be able to buy with your income?

    - More than today
    - The same
    - Less than today

- I don't know

19. Typically, which asset shows the largest fluctuations over time: a savings account, bonds, stocks, or cryptocurrencies?

    - Savings account
    - Bonds
    - Stocks
    - Cryptocurrency
    - I don't know

20. Which of the following statements describes the main function of the stock market?

    - The stock market helps predict stock earnings
    - The stock market results in an increase in stock prices
    - The stock market brings together people who want to buy stocks with those who want to sell them
    - None of the above
    - I don't know

C.4 Ambiguity/Risk Instructions

In each of the following activities, we will ask you to answer a series of questions. Each of these series of questions will be similar to the following list:

Row	**Option A: you win S/. 15 if some event X happens.**	**Option B: you win S/. 15 with a given probability.**
1.	Option A: you win S/. 15 if some event X happens.	Option B: you win S/. 15 with 0% probability.
2.	Option A: you win S/. 15 if some event X happens.	Option B: you win S/. 15 with 2.5% probability.
3.	Option A: you win S/. 15 if some event X happens.	Option B: you win S/. 15 with 5% probability.
4.	Option A: you win S/. 15 if some event X happens.	Option B: you win S/. 15 with 10% probability.
5.	Option A: you win S/. 15 if some event X happens.	Option B: you win S/. 15 with 20% probability.
6.	Option A: you win S/. 15 if some event X happens.	Option B: you win S/. 15 with 30% probability.
7.	Option A: you win S/. 15 if some event X happens.	Option B: you win S/. 15 with 40% probability.

Your Decisions

As you can see, in this example the table consists of 7 questions. In each question, you have to decide whether you prefer Option A (which pays S/. 15 if an event X has happened) or Option B (which pays S/. 15 with a given probability; moreover, this probability changes from row to row).

This event X could be, for example, whether it will rain in New York tomorrow.

You have to decide in each row whether you prefer Option A or Option B. In particular, you will choose one of the following options:

- I always prefer Option A
- I prefer Option B from row 1 onwards
- I prefer Option B from row 2 onwards
- I prefer Option B from row 3 onwards
- I prefer Option B from row 4 onwards
- I prefer Option B from row 5 onwards
- I prefer Option B from row 6 onwards
- I prefer Option B from row 7 onwards

In other words, you will decide from which row onwards you prefer Option B to Option A. This is because if you prefer Option B to Option A in row 5, you must also prefer Option B to Option A in row 6 (since in row 6 Option B pays S/. 15.00 with a higher probability than Option B in row 5, and Option A is always the same).

Important! If in row 5 you choose/prefer Option B to Option A, this means that you believe it will rain in New York tomorrow with a probability lower than (or equal to) 20%, since only then does Option B give you a higher chance of winning the S/. 15.00.

Your Payments

Your payment for each of these activities will be determined as follows. One of the questions in the list will be selected at random and with equal probability. Your decision in that question will determine your earnings. *Example.* Imagine that question 5 (row 5) is randomly selected. Now, if you selected Option A as your answer to this question, you will win S/. 15.00 if the event happens (it rains in New York tomorrow). If instead you selected Option B, you will win S/. 15.00 with a 15% probability.

Please note that although the following activities are similar, they are different. Each table will change in Options A and B and in the probabilities of winning S/. 15.00. So it is in your interest to read each of these activities carefully, as they could determine your payments in the experiment.

To continue, please type the number "40" in the box below.

C.5 Treatments

C.5.1 Financial Literacy

Training

Now you have the opportunity to earn an additional amount of money by choosing between different options in the following questions. Before making your decisions, we now give you information that could be relevant and that could help you make your decisions. Please read them carefully!

1. **What is a financial asset?** A financial asset is a financial instrument that can be traded on financial markets. Examples include:
    - **Stocks**: these are a capital investment that represents partial ownership of a corporation and gives the right to a share of the profits and assets of that corporation. An example of stocks is the shares of Banco de Crédito del Perú.
    - **Cryptocurrency**: this is a new form of digital asset based on a network distributed across a large number of computers. This decentralized structure allows them to exist outside the control of governments and central authorities. An example of cryptocurrency is Bitcoins.
2. **What is the price of a financial instrument?** The price of financial instruments reflects supply and demand factors in the market in which they operate. Therefore, over time, a particular financial asset may decrease in value, remain constant, or increase in value.

C.5.2 Probabilistic Literacy

Subjects received the same training as in the Financial Literacy treatment plus the following:

You now have the opportunity to earn an additional amount of money by choosing between different options in the following questions. Before you make your decisions, we will give you some information that may be relevant and could help you make your choices. Please read it carefully!

What are the chances that a financial instrument will increase or decrease in price over time? The probabilities of financial instruments increasing, remaining constant, or decreasing ultimately depend on the specific asset under consideration. **In particular, each probability is not necessarily equally likely.** For example, for a specific asset it might be more likely to increase in price, or vice versa (to drop in price). As a result, when making investment decisions, individuals must be aware of potential changes in the price of a financial instrument and form predictions about the probability of each possibility (increase or decrease in price).

C.6 Ambiguity Task

Subjects then faced 12 multiple-price lists like the following.

Bitcoin Return

For this activity, you have to consider the return of Bitcoin. In particular, the return of Bitcoin will be determined from the distribution of some past daily returns of the Bitcoin price. [If you would like more information on how a daily return is calculated, please read the box below the "Next" button.]

You will now choose between two options. Option A depends on a daily return of Bitcoin randomly drawn from the distribution of some past daily returns of the Bitcoin price.

- **Option A:** you win S/. 15 if the return of Bitcoin was strictly less than 0%.
- **Option B:** pays S/. 15 with a given probability. The probability increases from row to row. For example, in row 1 the probability is 0%, in row 2 the probability is 2.5%, and so on, until in row 15 the probability is 100%.

Indicate whether you prefer Option A or Option B. You do not have to make a choice in all 15 rows. If you select Option B in one row, your choice in all subsequent rows will automatically be set to Option B as well, and in all previous rows to Option A.

Therefore, you only have to select the row from which you prefer Option B onwards. It is also possible that you prefer Option A in every row. In that case, always select Option A.

Row	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with a given probability.
1.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 0% probability.
2.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 2.5% probability.
3.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 5% probability.
4.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 10% probability.
5.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 20% probability.
6.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 30% probability.
7.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 40% probability.
8.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 50% probability.
9.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 60% probability.
10.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 70% probability.
11.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 80% probability.
12.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 90% probability.
13.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 95% probability.
14.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 97.5% probability.
15.	Option A: you win S/. 15 if the return of Bitcoin was strictly less than 0%.	Option B: you win S/. 15 with 100% probability.

Now please decide your preferred option for each row.

The daily return of Bitcoin is obtained as the percentage increase or decrease in the closing price of Bitcoin on a given day compared with the closing price of the previous day. For example, if on November 4, 2020, the closing price is equal to 100 and on November 5 it is equal to 102, the daily return of the asset on November 5 will be +2.00%.

C.7 Risk Task

Subjects then faced 4 multiple-price lists like the following.

In this question you can win a prize depending on the outcome of a random coin toss. There is a 50% probability that the coin lands heads and a 50% probability that it lands tails. For each of the following 18 rows, choose whether you prefer Option A or Option B.

- **Option A:** pays S/. 15 if the coin lands heads (50% probability).
- **Option B:** pays a certain amount (with 100% probability). This amount increases across the rows of the table. For example, in row 1 the payment is S/. 0.00, in row 2 the payment is S/. 1.00, and so on, until in row 18 the payment is S/. 15.00.

Indicate whether you prefer Option A or Option B. You do not have to make a choice in all 18 rows. If you select Option B in a particular row, your choice in all subsequent rows will automatically be set to Option B as well, and in all previous rows to Option A.

Therefore, you only have to select the row from which you prefer Option B onwards. It is also possible that you prefer Option A in every row. In that case, always select Option A.

Row	Option A: you win S/. 15 if the coin lands heads.	Option B: you win the following amount with certainty.
1.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 0.00
2.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 1.00
3.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 2.00
4.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 3.00
5.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 4.00
6.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 4.50
7.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 5.00
8.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 5.50
9.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 6.00
10.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 6.50
11.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 7.00
12.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 7.50
13.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 8.00
14.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 9.00
15.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 10.00
16.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 11.00
17.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 12.50
18.	Option A: you win S/. 15 with 50% probability, you win S/. 0 with 50% probability.	Option B: certain payment of S/. 15.00

Now please decide your preferred option for each row.

C.8 Final Questionnaire

C.8.1 Financial Literacy Treatment

Read each question and choose the correct answer.

Before you made your first decisions (about Bitcoin and the shares of Banco de Crédito), we explained to you:

1. What is a financial asset?
2. What is the price of a financial instrument?

How useful were these concepts to you, on a scale from 1 to 5 (1: not useful at all; 5: very useful)?

- 1 (Not useful at all), 2, 3, 4, 5 (Very useful)

What is a stock?

- A certificate of ownership of part of a company
- A loan to a company
- The obligation to pay a company's debts
- None of the above
- I don't know

What is a Bitcoin?

- A new form of digital asset corresponding to ownership of a company
- A new form of digital asset based on a network distributed across a large number of computers
- A new form of digital asset that can only be used in Europe
- None of the above
- I don't know

C.8.2 Probabilistic Literacy Treatment

Read each question and choose the correct answer.

Before you made your first decisions (about Bitcoin and the shares of Banco de Crédito), we explained to you:

- What are the chances that a financial instrument will increase or decrease in price over time?

How useful was this concept to you, on a scale from 1 to 5 (1: not useful at all; 5: very useful)?

- 1 (Not useful at all), 2, 3, 4, 5 (Very useful)

What is a stock?

- A certificate of ownership of part of a company
- A loan to a company
- The obligation to pay a company's debts
- None of the above
- I don't know

What is a Bitcoin?

- A new form of digital asset corresponding to ownership of a company
- A new form of digital asset based on a network distributed across a large number of computers
- A new form of digital asset that can only be used in Europe
- None of the above
- I don't know

C.8.3 Control

Read each question and choose the correct answer.

What is a stock?

- A certificate of ownership of part of a company

- A loan to a company
- The obligation to pay a company's debts
- None of the above
- I don't know

What is a Bitcoin?

- A new form of digital asset corresponding to ownership of a company
- A new form of digital asset based on a network distributed across a large number of computers
- A new form of digital asset that can only be used in Europe
- None of the above
- I don't know

C.9 End of the Experiment

Please continue with the last two pages of the experiment.

C.9.1 Payment Page

In this experiment, we will pay you S/. 5.00 for your participation plus your earnings in one part of the experiment that is randomly chosen by the computer.

In the activity randomly chosen by the computer, you have earned S/. [payoff] in total.

We will therefore pay you S/. 5.00 + S/. [payoff] for your participation.

C.9.2 Final Page

To proceed with your payment, click on the link below. This link will redirect you to a form where you can provide the information needed for you to receive your payment.

`http://bit.ly/PUCP_LEEX_InfoPago`

In addition, you will need to enter your participant label:
[participant ID]

The data you provide in that form will not be combined with the data from the experiment and will not affect the amount of your payment.

If you have any problems during the experiment or if you have any questions, write to this email address: `leex@pucp.edu.pe`, or call the following phone number: +51 941 475 026.

Thank you again!

LEEX Team – PUCP